ANNEXURE - I

Online Medical Diagnosis System (OMDS)

By

Pratham Buddhadev - 92110133003

Under the guidance of
Prof. Harikesh Chauhan

A Project Submitted to

Marwadi University in Partial Fulfillment of the Requirements for the Bachelor of Technology in Information and Communication Technology

April 2023

MARWADI UNIVERSITY
Rajkot-Morbi Road, At & Po. Gauridad,
Rajkot-360003, Gujarat, India.

ANNEXURE – II

CERTIFICATE

This is to certify that research/project work embodied in this dissertation titled "Online Medical Diagnosis System" was carried out by **Pratham Buddhadev** at **Marwadi University** for partial fulfillment of **Bachelor of Technology** in **Information and Communication Technology** to be awarded by Marwadi University. This research/project work has been carried out under my guidance and supervision and it is up to my satisfaction.

Date:

Place:

Prof. Harikesh Chauhan
Project Guide

Prof. Chandrasinh Parmar
Head of Dept.

Dr. Rajendrasinh Jadeja
Principal/Dean

Seal of Institute

COMPLIANCE CERTIFICATE

This is to certify that research/project work embodied in this dissertation titled "Online Medical Diagnosis System" was carried out by **Pratham Buddhadev - 92110133003** at **Marwadi University** for partial fulfillment **Bachelor of Technology** to be awarded by Marwadi University. He/ She has complied to the comments given during Review I, Review II, Review III by Reviewer to my satisfaction.

Date:

Place:

Pratham Buddhadev | **Prof. Harikesh Chauhan**

92110133003 | **Project Guide**

<u>**PROJECT/PROJECT APPROVAL CERTIFICATE**</u>

This is to certify that research/project work embodied in this dissertation titled “Online Diagnosis Management System” was carried out by **Pratham Buddhadev - 92110133003** at **Marwadi University** is approved for the **Bachelor of Technology** in **Information and Communication Technology** by Marwadi University.

Date:

Place:

Examiner’s Sign and Name:

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( ) ( )

ANNEXURE - IV

UNDERTAKING ABOUT ORIGINALITY OF WORK

We hereby certify that we are the sole authors of this project/project work and that neither any part of this project nor the whole of the project has been submitted for a degree to any other University or Institution.

We certify that, to the best of our knowledge, the current project/project work does not infringe upon anyone's copyright nor violate any proprietary rights and that any ideas, techniques, quotations or any other material from the work of other people included in our project/project work, published or otherwise, are fully acknowledged in accordance with the standard referencing practices. Furthermore, to the extent that we have included copyrighted material that surpasses the boundary of fair dealing within the meaning of the Indian Copyright (Amendment) Act 2012, we certify that we have obtained a written permission from the copyright owner(s) to include such material(s) in the current project and have included copies of such copyright clearances to our appendix.

We declare that this is a true copy of project/project work, including any final revisions, as approved by project/project work review committee.

We have checked write up of the present project/project work using anti-plagiarism database and it is in allowable limit. Even though later on in case of any complaint pertaining of plagiarism, we are sole responsible for the same and we understand that as per UGC norms, University can even revoke Degree name conferred to the student submitting this project.

Date:

Pratham Buddhadev **Prof. Harikesh Chauhan**

92110133003 **Project Guide**

TABLE OF CONTENTS

1. Introduction

1.1 Literature Review

[1] Automatized Medical Chatbot (Medibot)
Authors: Prakhar Srivastava, Nishant Singh

This paper explores the impact of Medibot, an automatized medical chatbot System presented at the 2020 PARC conference. Srivastava and Singh discuss how Medibot leverages AI and IoT to provide accessible healthcare assistance. They examine its integration with renewable energy systems and IoT devices, its benefits, challenges, and its potential to improve healthcare delivery. The authors advocate for further research in this area to advance patient-centric care.

[2] Chatbot for Healthcare System using AI
Authors: Lekha Athota, Vinod Kumar Shukla, Nitin Pandey, Ajay Rana

This paper introduces the concept of a medical chatbot utilizing Artificial Intelligence (AI) to offer preliminary disease diagnosis and basic medical information, thereby reducing healthcare costs and improving accessibility to medical expertise. The chatbot interacts with users using natural language processing, storing data in a database to identify keywords and formulate responses. Utilizing techniques like n-gram, TFIDF, and cosine similarity, the chatbot ranks and retrieves relevant information to answer user queries. Additionally, an expert program manages queries beyond the chatbot's capabilities. Published in the 2020 8th International Conference on Reliability, Infocom Technologies, and Optimization (ICRITO), the paper underscores the potential of AI-driven chatbots to enhance healthcare accessibility and affordability.

[3] The Potential of Chatbots: Analysis of Chatbot Conversation
Authors: Mubashra Akhtar, Julia Neidhardt, Hannes Werthner

This paper explores the utilization of chatbots in improving customer relationships, focusing on their role within telecommunication companies. Analyzing chat conversations between customers and a telecommunication company's chatbot, the study aims to identify users' topics of interest and assess user satisfaction. Through text mining techniques, the paper interprets chat conversations as sequences of events, revealing valuable insights into user behavior. Findings indicate that users often leave the conversation quickly if their queries are not promptly resolved, highlighting the importance of real-time responses. Additionally, the study identifies recurring topics in conversations, suggesting the need for companies to analyze data comprehensively to

understand customer needs better. The paper concludes that personalized service and real-time feedback are key to improving customer satisfaction in chatbot interactions. Published in the 2019 IEEE 21st Conference on Business Informatics

(CBI), the paper underscores the significance of leveraging chatbot data for enhancing customer experience in telecommunication services.

1.2 Problem Summary and Introduction

The problem arises when we need doctors help immediately, but they are not available due to some reason like Covid-19. The health facilities are far-fetched. So, due to the unavailability of medical team and doctors people sometimes get misdiagnosed. The procedure for diagnosing also consume a big amount of time although the patients come's with a minor problem that they can settle it by themselves if they have the right help.

The Medical Diagnosis system is a system that helps the user to get the appropriate information needed regarding overall health conditions and in developing a better lifestyle. This system is split into three main parts that comprise of Patient's Registration and Doctor, Diagnosis and Treatment and Health Monitor and Tips.

1.3 Purpose

The primary purpose of the Medical Diagnosis system is to offer immediate medical guidance and support to users, ensuring timely intervention in health-related emergencies. By providing accurate information and guidance, the system aims to reduce the risk of misdiagnosis, which can lead to adverse health outcomes and unnecessary medical interventions. The system empowers users by equipping them with the necessary information to make informed decisions about their health and well-being. It encourages self-care and proactive health management.

Through health monitoring features and lifestyle tips, the system promotes a holistic approach to healthcare, encouraging users to adopt healthier habits and practices for improved overall well-being. By automating certain aspects of the diagnosis process and providing remote assistance, the system contributes to the overall efficiency of healthcare delivery, especially in resource-constrained environments. The system complements the efforts of healthcare professionals by triaging patient inquiries, providing preliminary assessments, and facilitating more efficient use of medical resources.

1.4 Scope

- The Medical Diagnosis system aims to bridge the gap in healthcare accessibility by providing immediate assistance to users, especially in situations where medical professionals are unavailable or health facilities are distant.

- It focuses on addressing urgent medical needs, particularly during crises like the Covid-19 pandemic, where access to healthcare services may be restricted.

- The system aims to streamline the diagnosis process, minimizing the time and resources required for accurate medical assessments. It caters to users presenting with minor health issues that can be managed with timely assistance.

- By offering comprehensive information on overall health conditions and lifestyle improvements, the system encourages users to adopt healthier habits and proactive healthcare measures.

1.5 Aim and Objectives

- Improve access to healthcare services, especially in remote or crisis-affected areas.
- Offer immediate medical guidance and support for urgent health needs.
- Minimize the risk of incorrect diagnoses, ensuring better health outcomes.
- Equip individuals with knowledge and resources for informed healthcare decision-making.
- Increase awareness of health issues and encourage healthier lifestyles.
- Develop an intuitive interface for easy navigation and access to medical information.
- Utilize AI algorithms for accurate medical guidance and diagnosis assistance.
- Include features for real-time tracking of vital health metrics.
- Provide tailored health advice based on user data and preferences.
- Implement robust security protocols to safeguard user privacy.
- Conduct thorough testing to refine system usability based on user input.
- Partner with healthcare experts to ensure the system meets medical standards.
- Employ tactics to maintain user interest and adherence to the system.

1.6 Problem Specifications

- Limited Healthcare Accessibility
- Lack of Immediate Medical Assistance
- Risk of Misdiagnosis
- Time-Consuming Diagnosis Procedures
- Need for Holistic Solutions

1.7 Technologies used for Development

- Python Flask was employed for backend development, providing a lightweight and efficient framework for building web applications. This allowed for rapid development and easy integration with other technologies.

- JavaScript played a crucial role in enhancing the system's interactivity and dynamic functionality. Leveraging JavaScript, the system was able to deliver real-time updates, interactive features, and seamless user interactions, enhancing the overall user experience.

- Additionally, PHP was utilized for certain server-side functionalities, complementing the Python Flask backend. PHP's versatility and compatibility with databases such as MySQL and PostgreSQL allowed for efficient data management and processing.

- Firebase database served as the backbone for storing and managing user data securely. Its real-time database capabilities ensured that users received up-to-date information and personalized recommendations in a timely manner.

- XAMPP, a cross-platform web server solution, facilitated local development and testing of the system. This allowed for thorough testing and debugging before deployment, ensuring a stable and reliable system.

- Moreover, Kotlin was utilized for mobile development, enabling the creation of a companion mobile application for seamless access to the Medical Diagnosis system on-the-go. This extended the system's reach and accessibility to a wider user base.

1.8 Materials/Tools required

[1] Hardware requirement of the project

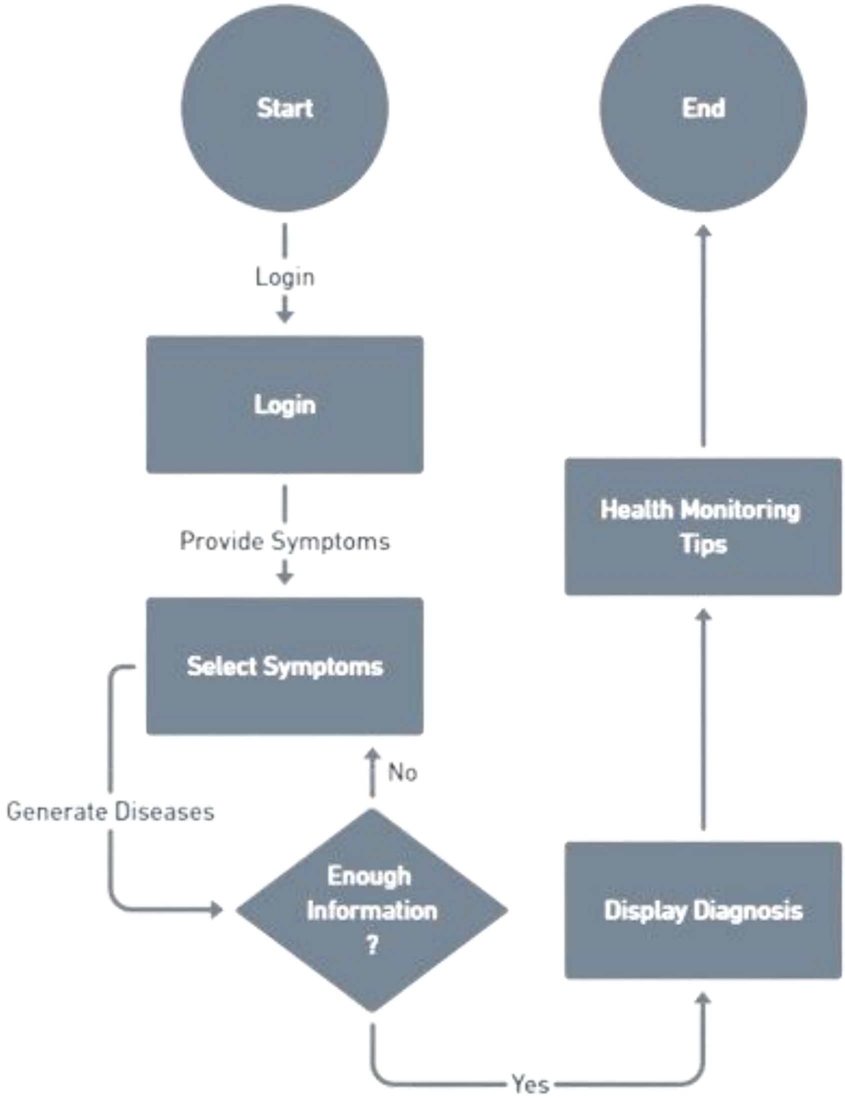
1. Laptop/desktop – For development
2. Internet – For fetching data and testing

[2] Software requirement of the project

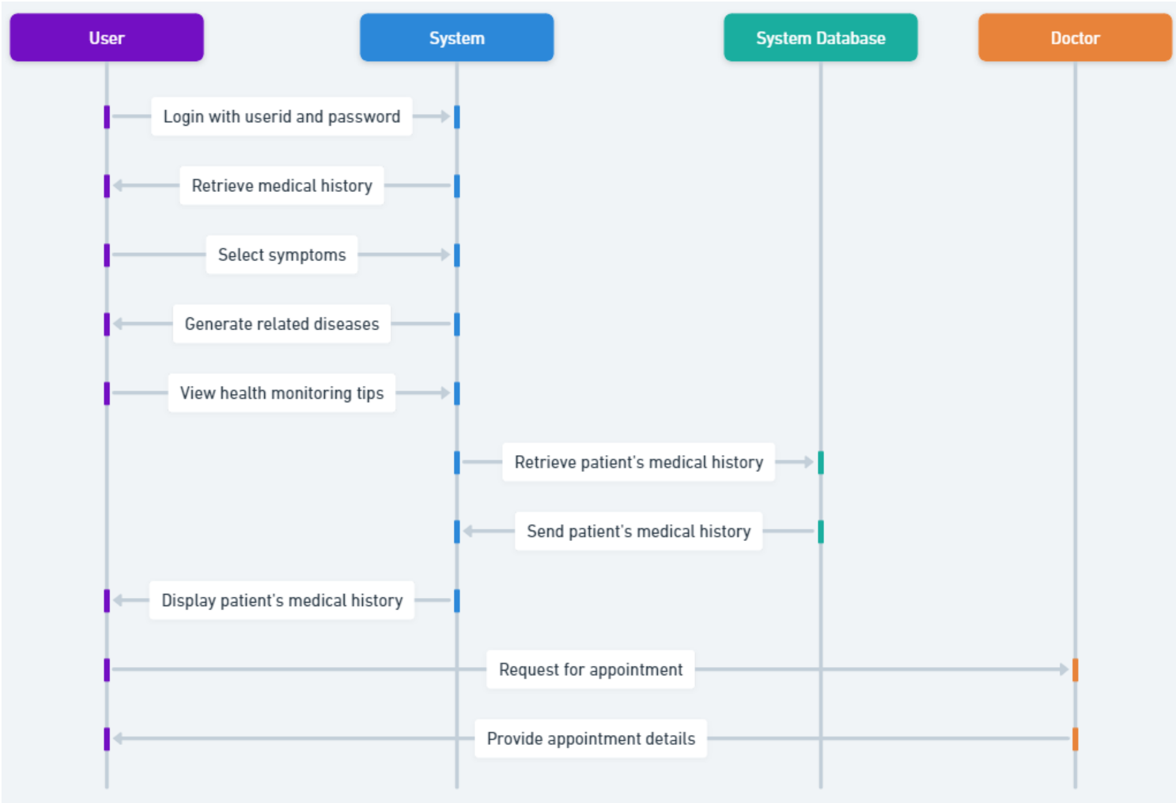
1. Visual Studio Code – Editor for development
2. Android Studio – Editor for development
3. Mac OS / Windows OS Platform
4. GitHub – To manage and collaborate on code effectively

2. Analysis, Design Methodology, and Implementation Strategy

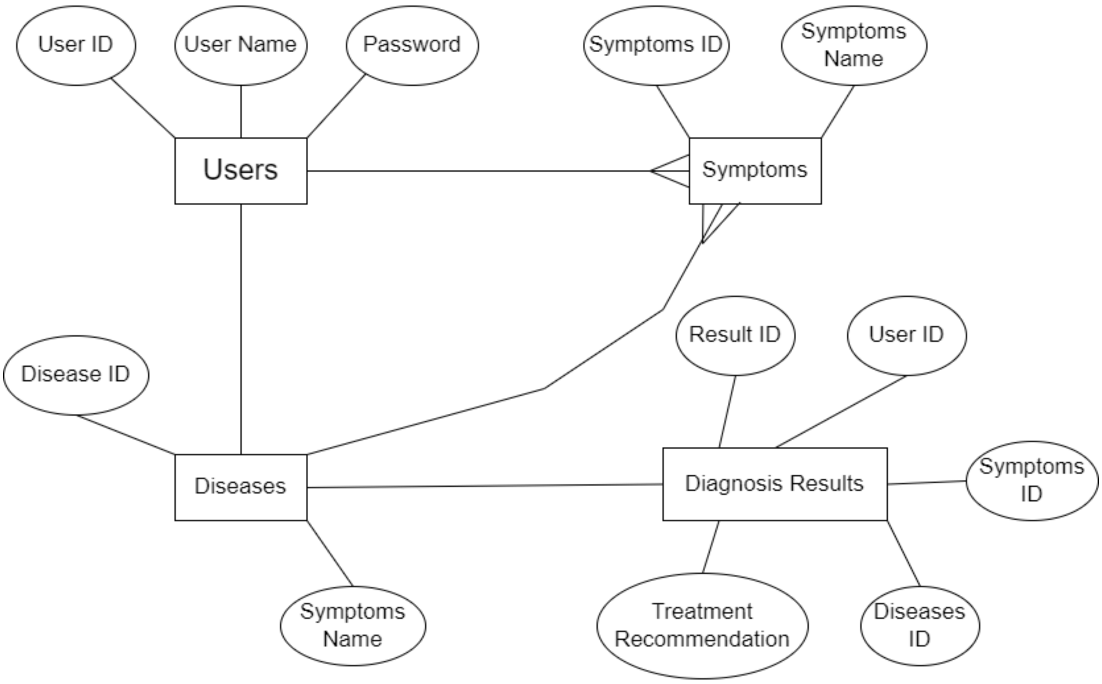
2.1 Flow Chart



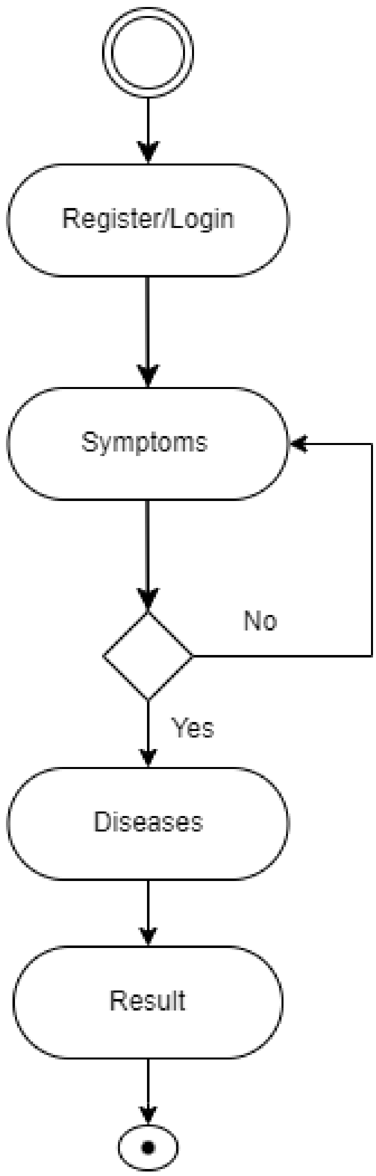
2.2 Sequence Diagram



2.3 ER Diagram



2.4 Activity Diagram



3. Implementation

3.1 Implemented Functionality

For the implementation of the Online Medical Diagnosis system, a comprehensive set of functionalities was developed to ensure seamless operation and user interaction. The system was built using a combination of Python Flask, Firebase database, PHP, and JavaScript, enabling robust functionality and efficient data management.

- **Backend Development with Python Flask:** The system's backend was developed using Python Flask, a lightweight and efficient web framework. Flask provided a solid foundation for building RESTful APIs and handling server-side logic.

- **Firebase Database Integration:** Firebase database was utilized for storing and managing user data securely. Its real-time database capabilities ensured that users received up-to-date information and personalized recommendations in a timely manner.

- **PHP for Server-Side Processing:** Certain server-side functionalities were implemented using PHP, complementing the Python Flask backend. PHP's versatility and compatibility with databases such as MySQL and PostgreSQL allowed for efficient data management and processing.

- **JavaScript for Dynamic Frontend:** JavaScript played a crucial role in enhancing the system's interactivity and dynamic functionality. Leveraging JavaScript, the system was able to deliver real-time updates, interactive features, and seamless user interactions, enhancing the overall user experience.

- **Mobile Application Development with Kotlin:** Kotlin was utilized for mobile development, enabling the creation of a companion mobile application for seamless access to the Online Medical Diagnosis system on-the-go. This extended the system's reach and accessibility to a wider user base.

3.2 Results and Reports

- **Successful Development:** The implementation of the Medical Diagnosis system has been deemed successful, with all planned features and functionalities executed according to project specifications and requirements.

- **Optimized Performance:** Extensive local testing has demonstrated high performance and speed of the system, attributed to the adherence to best practices for optimization and caching techniques. This ensures an efficient and responsive user experience.

- **Efficient Data Management:** The custom functionalities integrated into the system have significantly enhanced the management of medical data. The implementation of a custom REST API endpoint and Firebase database integration allows for seamless data access and manipulation. Additionally, the use of Python Flask for backend development ensures efficient handling of server-side operations.

- **Granular Control and Accessibility:** Custom user roles and permissions have been implemented, enabling granular control over user access and data management. The inclusion of Firebase database facilitates real-time data updates and synchronization, ensuring users have access to the most up-to-date information.

- **Improved User Experience:** Feedback from beta testers and stakeholders has been overwhelmingly positive, highlighting the system's functionality and ease of use. This positive reception indicates that the Online Medical Diagnosis system is effectively meeting the needs and expectations of its intended users, promising value and utility for healthcare professionals and patients alike.

3.3 Snapshots

Figure 3.3.1 Choose your Category

The above figure shows choose your category if you are a doctor then choose doctor or else User.

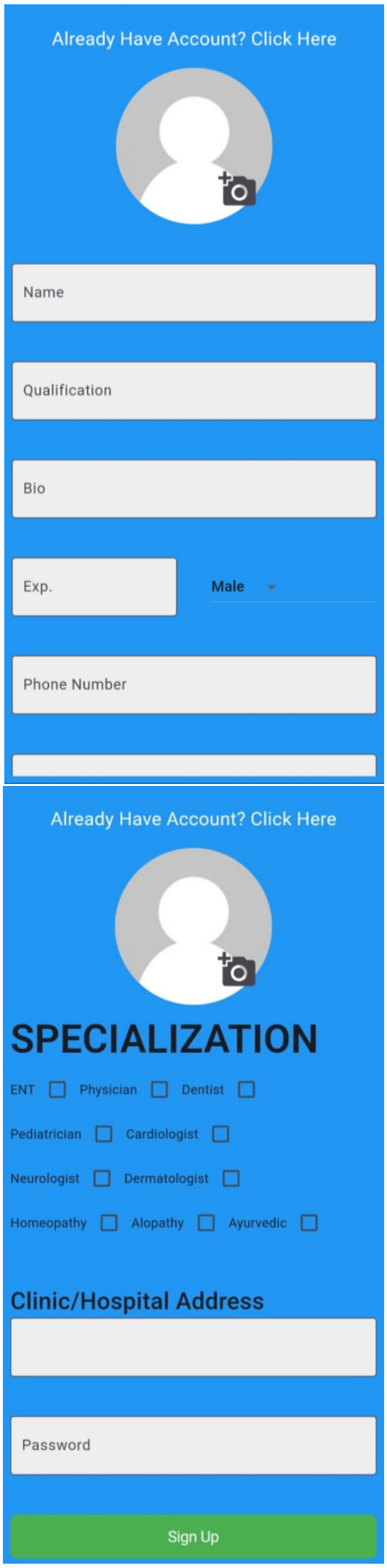


Figure 3.3.2 Doctor Registration page

The above figure shows Registration Page of Doctor

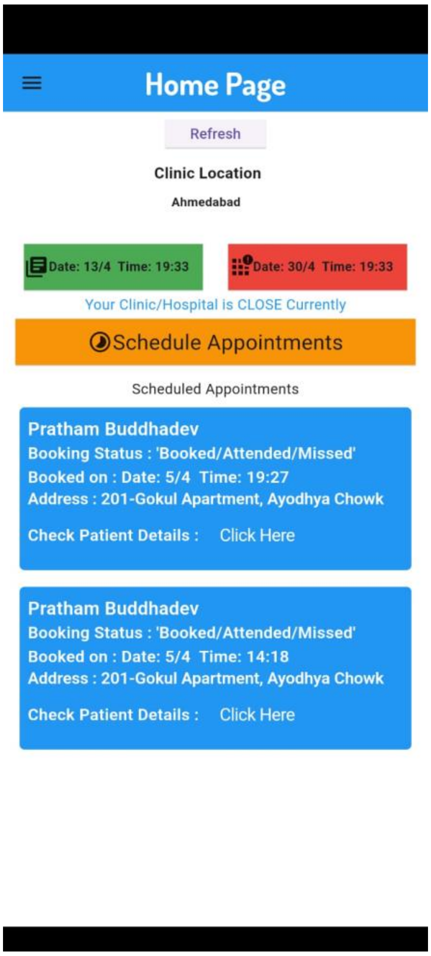


Figure 3.3.3 Doctor's Home Page/Dashboard

The above figure shows Doctor's Home/Dashboard Page

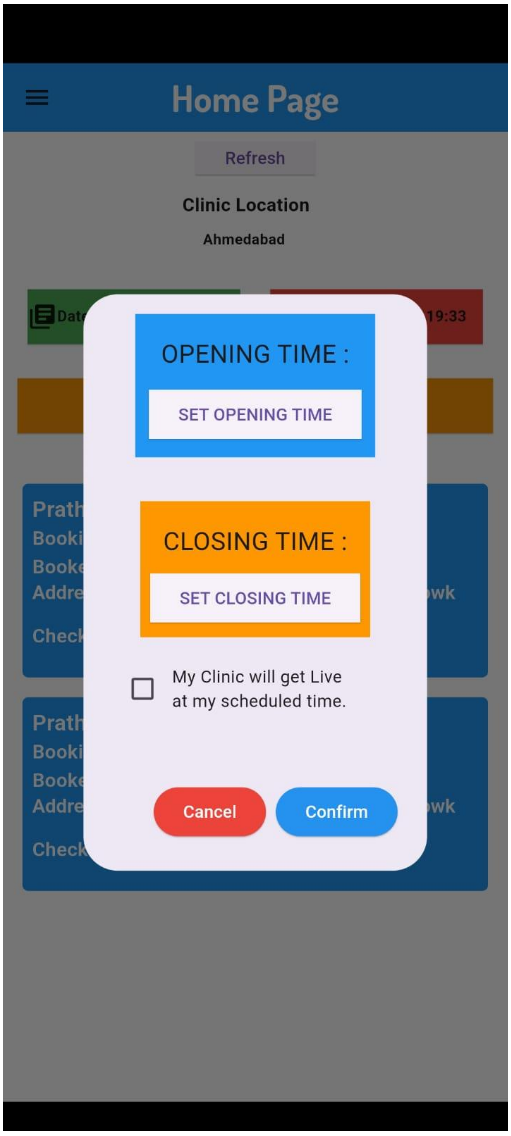


Figure 3.3.4 Set Appointment Schedule Time

The above figure shows Doctor's can set appointment time

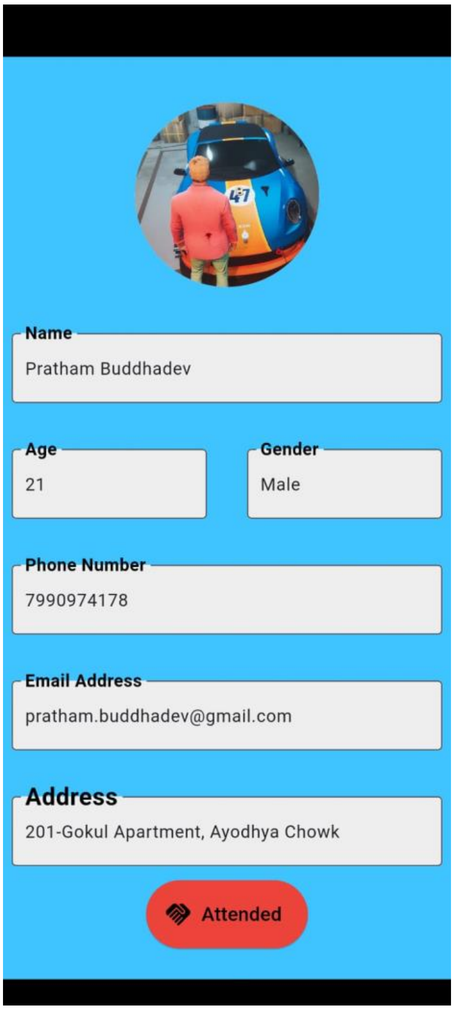


Figure 3.3.5 Doctor's Attending Patient Page

The above figure shows Doctor has attend a Patient

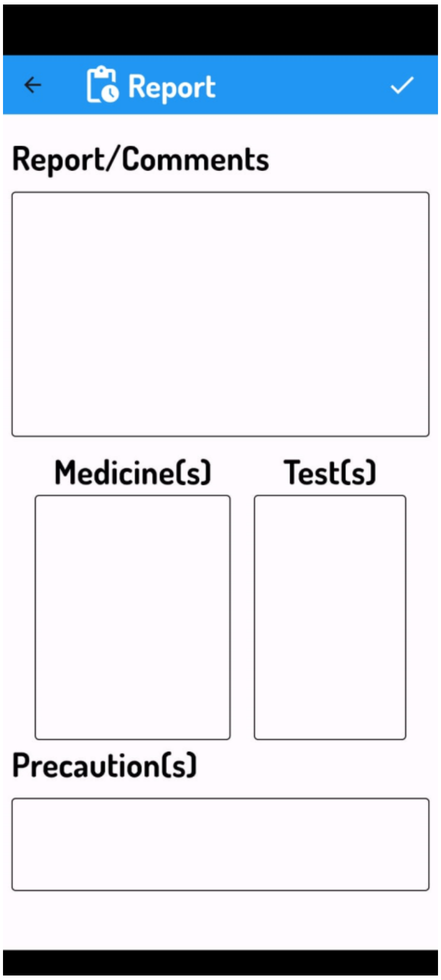


Figure 3.3.6 Write Report of the Patient

The above figure shows if doctor attend patient after that doctor have to write's report of that patient.

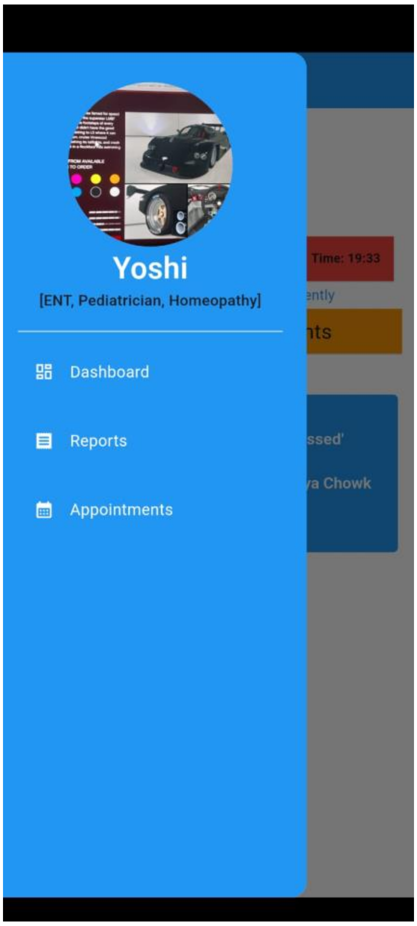


Figure 3.3.7 Doctor's Drawn Menu Bar

The above figure shows doctor side menu bar.

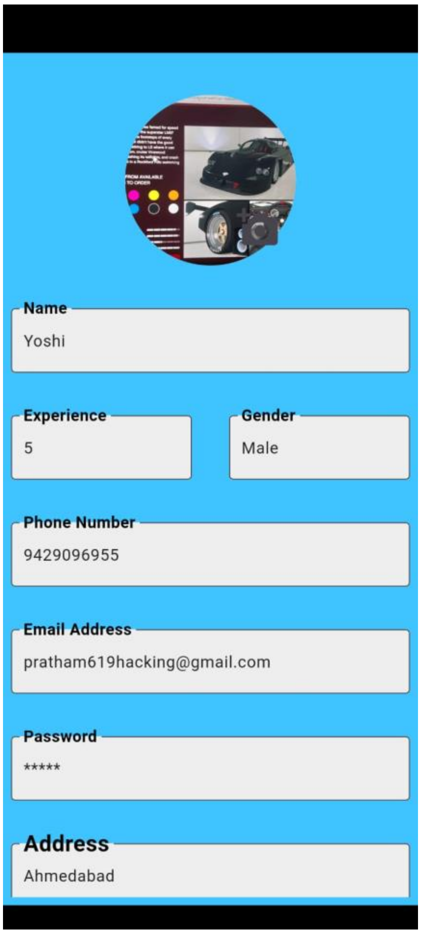


Figure 3.3.8 Doctor's Profile

The above figure shows Doctor's Profile

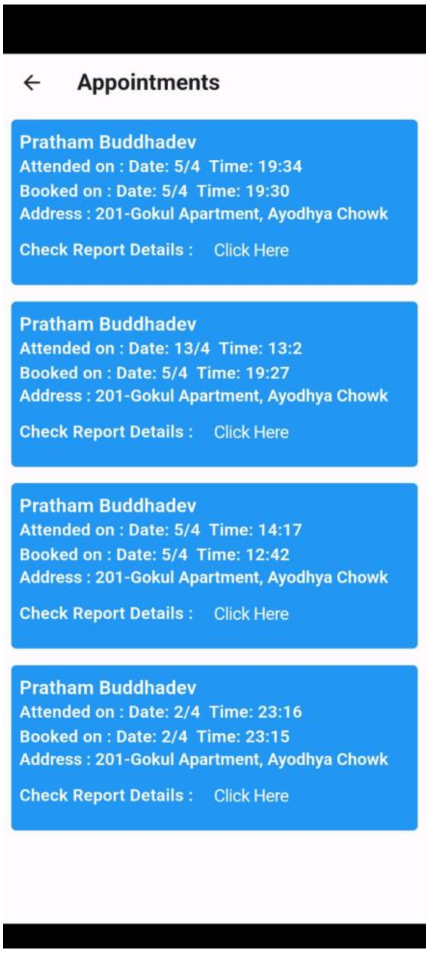


Figure 3.3.9 Shows reports of attended Patient’s

The above figure shows all the reports of the attended patient’s

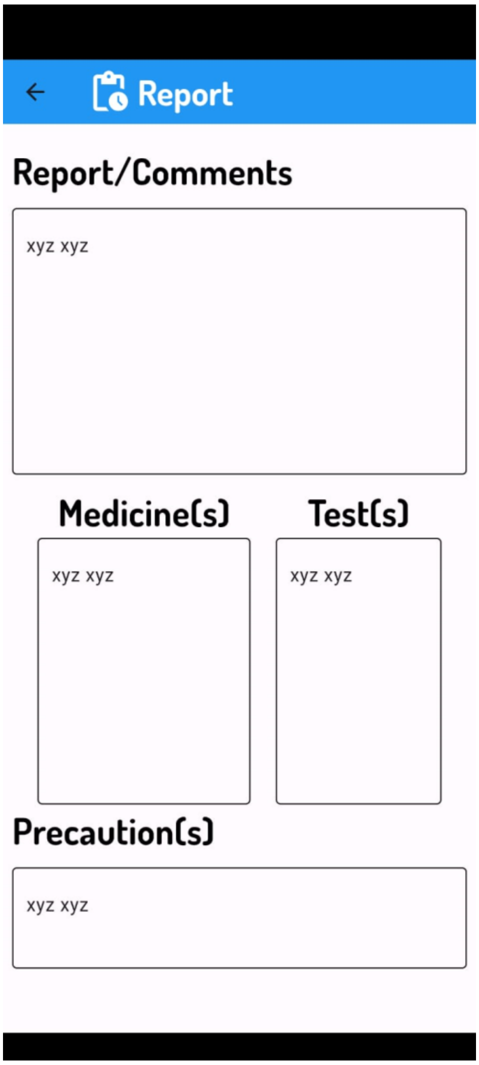


Figure 3.3.10 Reports

The above figure shows reports of the patients

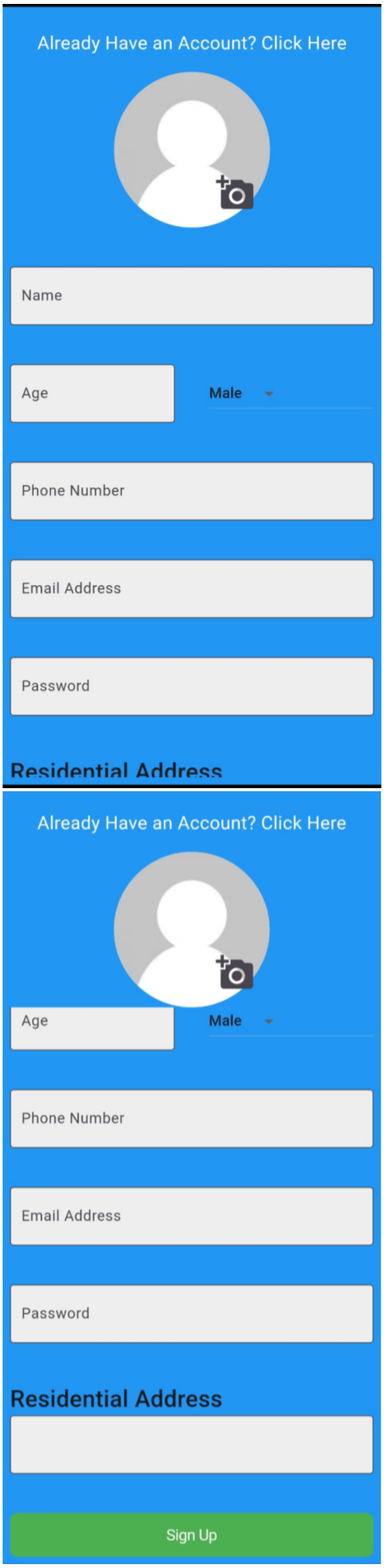


Figure 3.3.11 Patient's side Registration

The above figure shows Patient's side registration page

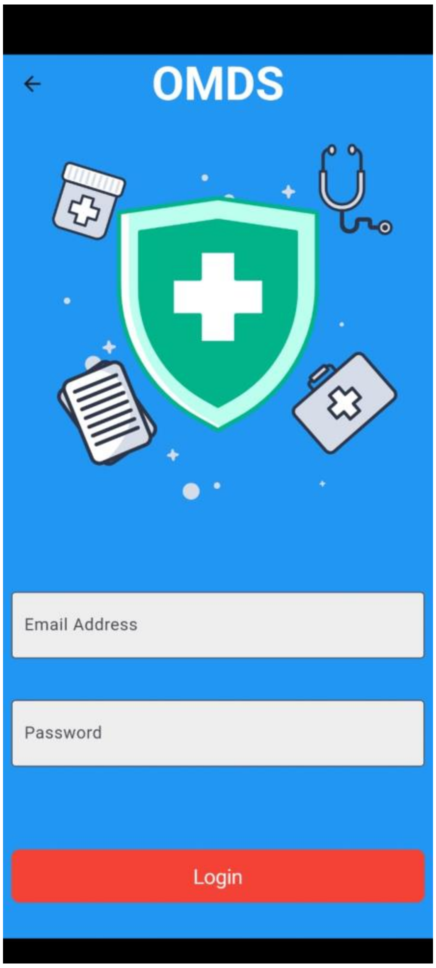


Figure 3.3.12 Patient's Login Page

The above figure shows login side of Patient's

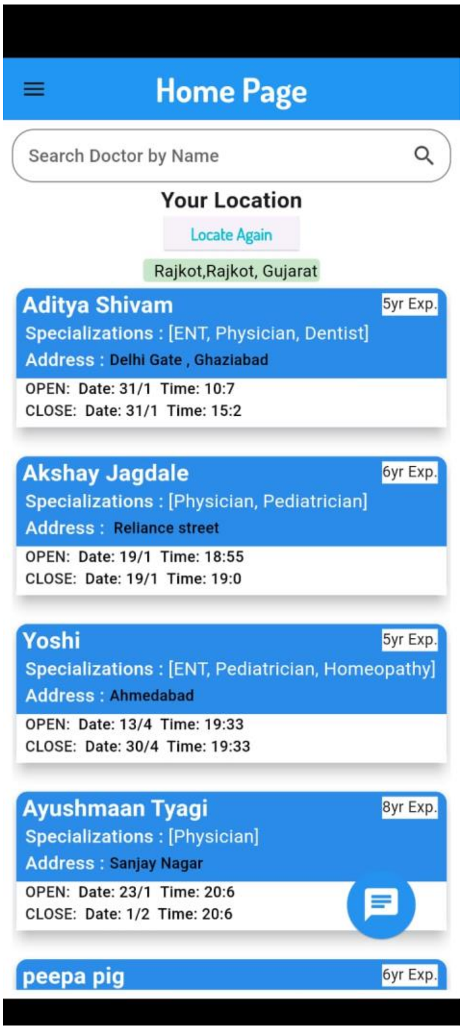


Figure 3.3.13 Patient's side Dashboard

The above figure shows Patients can see doctor's information from their home page and also take appointment of doctor for them. There is search bar for to find doctors.

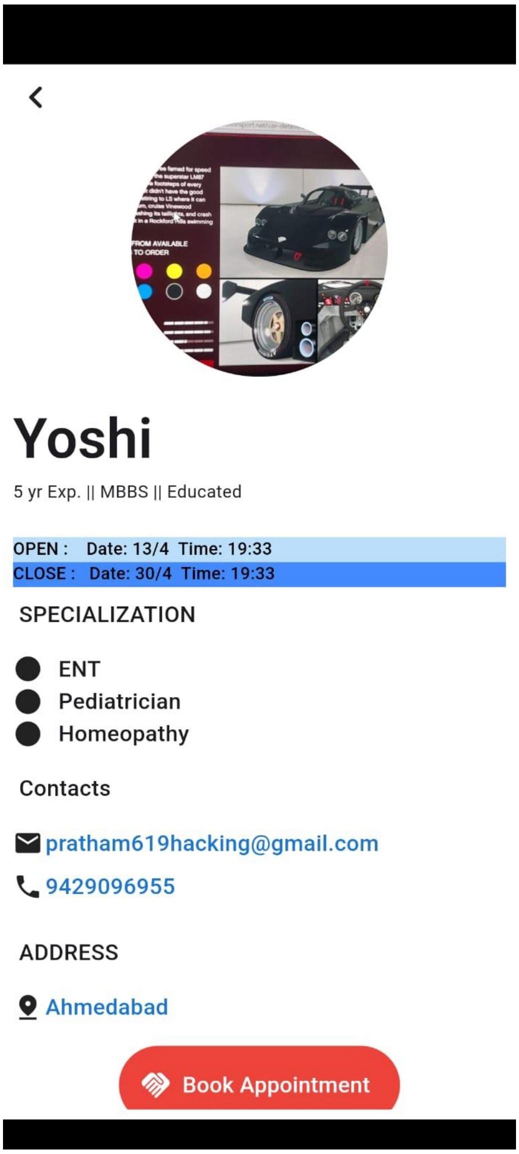


Figure 3.3.14 Doctor's Profile on User side

The above figure shows Patient can also see Doctors Profile form their Dashboard

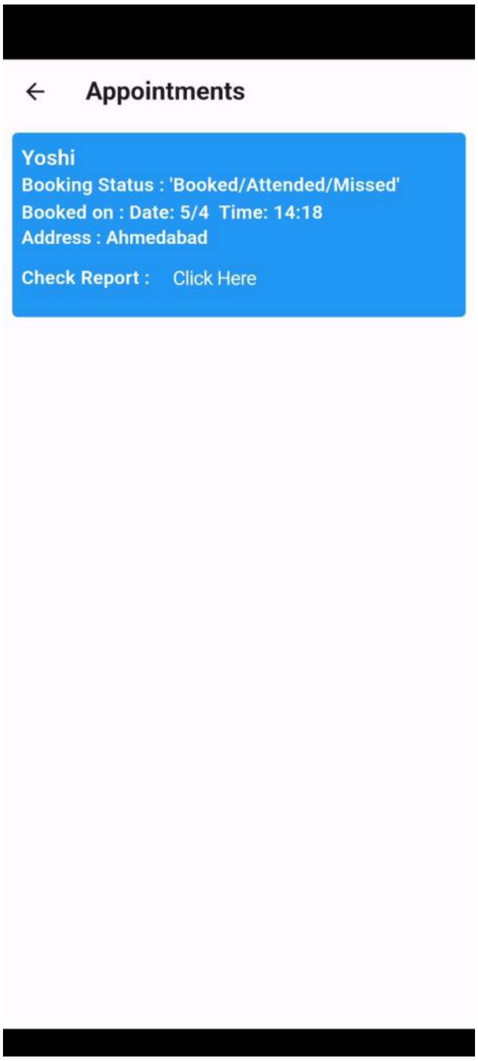


Figure 3.3.15 Patient Appointment Schedule

The above figure shows Doctors appointment Schedule details to user

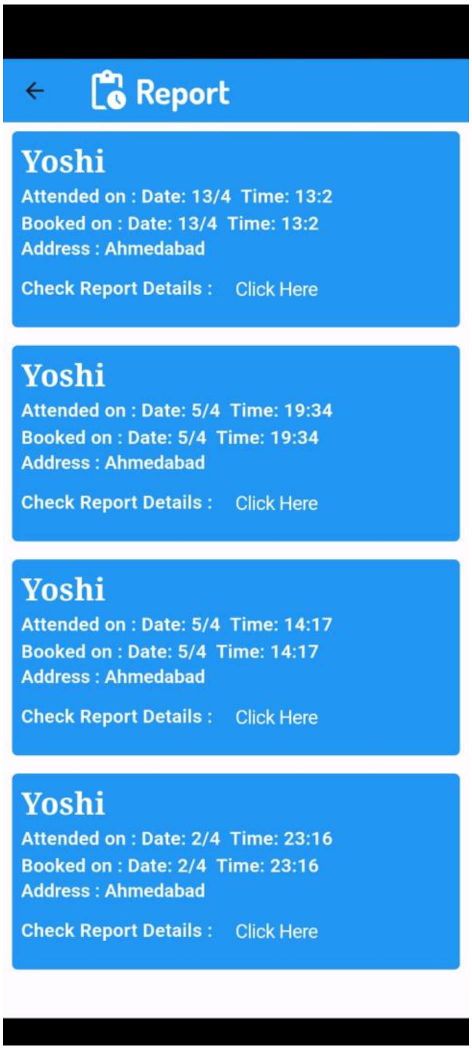


Figure 3.3.16 Patients reports list out

The above figure shows Patients reports list out

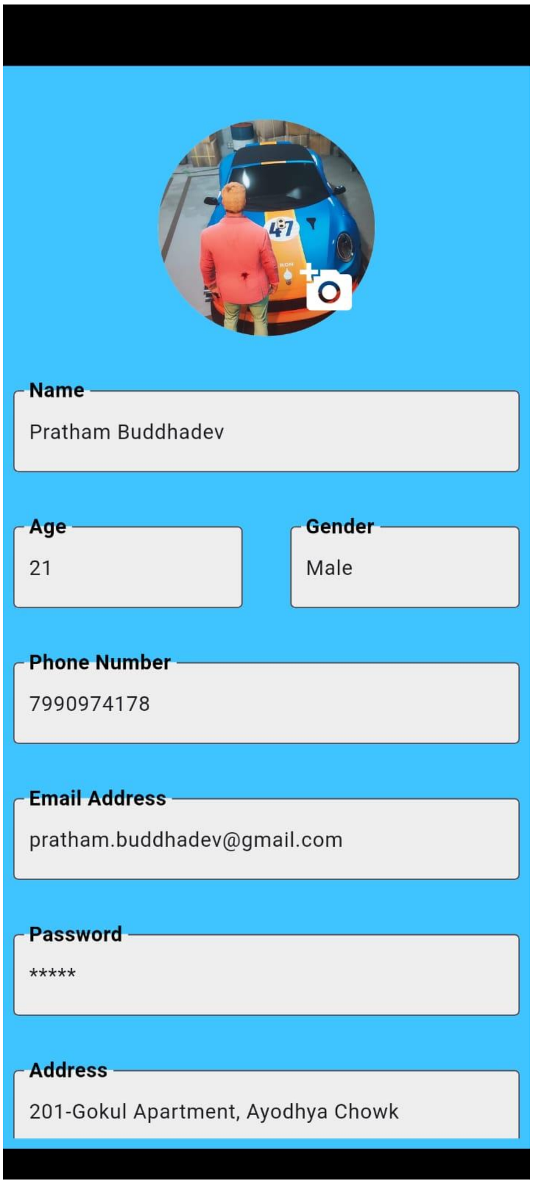


Figure 3.3.17 Patient's Profile

The above figure shows Patient's Profile

Figure 3.3.18 AI Healthcare Chatbot

The above figure shows Patients can chat with AI for getting precautions from the diseases.

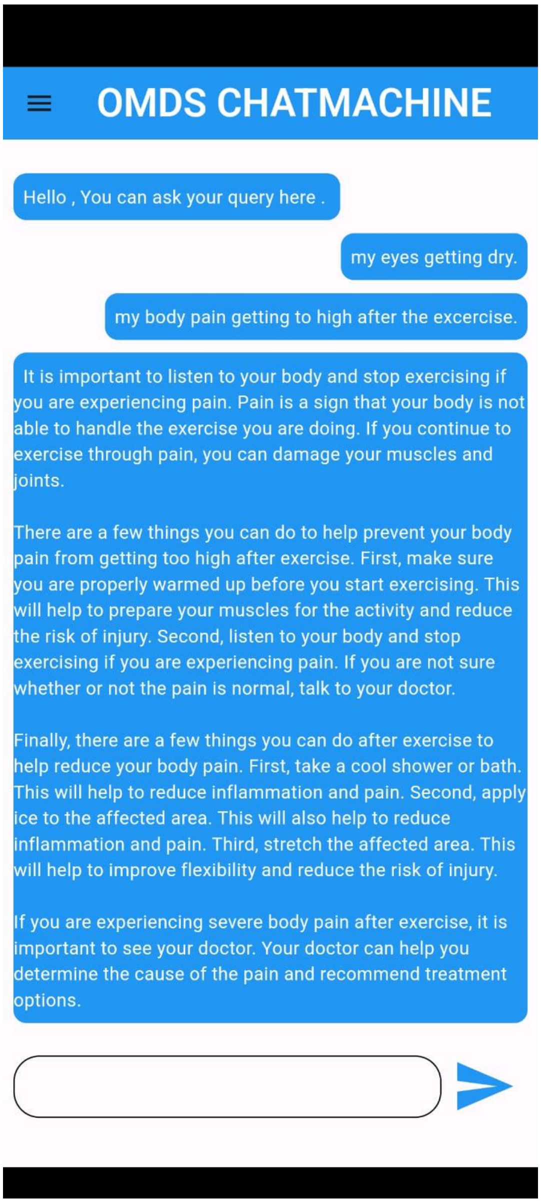


Figure 3.3.19 Final Result

3.4 Testing and Verification

- **Functionality Testing:** Rigorous testing has been conducted on all custom functionalities integrated into the Medical Diagnosis system, ensuring that features such as the custom REST API endpoint and user roles function as intended and meet project requirements.

- **Security Testing:** The system underwent comprehensive security testing to identify and mitigate potential vulnerabilities, including measures to prevent cross-site scripting (XSS) and SQL injection attacks. Implementation of SSL encryption, secure password policies, and CSRF protection ensures robust security measures are in place.

- **Performance Testing:** Performance testing was carried out to assess the system's ability to handle high traffic loads efficiently. Optimization techniques were applied to enhance loading speed and responsiveness across various devices, resulting in an average lighthouse score of 85% for performance.

- **Compatibility Testing:** The system underwent thorough compatibility testing to ensure seamless functionality across different browsers and devices. Compatibility checks were performed to verify proper display and functionality on all major platforms, ensuring a consistent user experience.

- **Accessibility Testing:** The system was tested for accessibility to ensure usability by individuals with disabilities. Measures such as providing alt text for images and keyboard navigation were implemented to enhance accessibility, resulting in an average lighthouse score of 90% for accessibility. Consideration was given to users in minority demographics during the development process.

4. Conclusion

4.1 Summary of the results

The implementation of the Medical Diagnosis system has resulted in a dynamic and user-friendly platform tailored for healthcare management. The system incorporates various functionalities, including a Python Flask backend, Firebase database integration, PHP for server-side processing, and JavaScript for dynamic frontend interactions.

Key features of the system include real-time data updates, personalized recommendations, and seamless integration with mobile applications using Kotlin. Through rigorous testing and optimization measures, the system demonstrates high performance, security, and compatibility across different devices and browsers.

4.2 Advantages of your work/results/methodologies

- **Comprehensive Healthcare Information:** The system offers a wealth of information about various health conditions, treatments, and lifestyle recommendations. This comprehensive data empowers users to make informed decisions about their health and well-being, benefiting both individuals seeking medical advice and healthcare professionals.

- **Efficient Organization and Categorization:** Custom functionalities such as personalized user profiles, medical histories, and treatment plans facilitate efficient organization and categorization of patient data. This streamlines healthcare management processes, enabling healthcare providers to access relevant information quickly and effectively.

- **User-Friendly Interface:** The intuitive user interface of the system ensures ease of navigation and accessibility for users of all levels of technical proficiency. This enhances user satisfaction and encourages continued engagement with the system, ultimately leading to better health outcomes.

- **Custom REST API Endpoint:** The implementation of a custom REST API endpoint enables seamless integration with third-party applications and services, facilitating data exchange and interoperability. This extends the system's functionality and versatility, enabling users to leverage additional tools and resources for their healthcare needs.

- **Real-Time Data Updates:** The system provides real-time updates on medical information, treatment protocols, and healthcare resources. This ensures that users have access to the latest and most relevant information, promoting proactive healthcare management and informed decision-making.

4.3 Scope of Future work

- **Enhanced Search Functionality:** Future enhancements could focus on improving the search functionality of the system by implementing advanced search features. This could include the ability to search by various criteria such as symptoms, medical conditions, treatment options, and more, providing users with a more tailored and efficient search experience.

- **Integration with External Healthcare Databases:** Integrating the Medical Diagnosis system with external healthcare databases and resources, such as medical journals, clinical trials databases, and patient forums, can expand the system's data coverage and provide users with access to a broader range of medical information and resources. This integration can enhance the system's capabilities and provide users with more comprehensive and up-to-date information.

- **Personalized Recommendations and Decision Support:** Incorporating machine learning algorithms into the system can enable personalized recommendations for medical treatments, lifestyle interventions, and preventive care measures based on individual user characteristics, medical history, and preferences. This personalized approach can enhance user engagement and satisfaction while promoting proactive healthcare management.

- **Advanced Analytics and Reporting:** Implementing advanced analytics and reporting functionalities can provide valuable insights into user behaviour, system usage patterns, and effectiveness of medical interventions. This data-driven approach can help identify trends, evaluate the impact of different interventions, and inform decision-making processes for future system enhancements and developments.

4.4 Unique Features of Project

- **Custom REST API Endpoint:** The system boasts a custom REST API endpoint, enabling seamless interaction with its data from any platform. This unique feature facilitates easy integration with third-party applications and services, enhancing the system's versatility and accessibility.

- **Advanced Analytics and Reporting:** Implementing advanced analytics and reporting functionalities can provide valuable insights into user behavior, system usage patterns, and effectiveness of medical interventions. This data-driven approach can help identify trends, evaluate the impact of different interventions, and inform decision-making processes for future system enhancements and developments.

- **Personalized Recommendations and Decision Support:** Incorporating machine learning algorithms into the system can enable personalized recommendations for medical treatments, lifestyle interventions, and preventive care measures based on individual user characteristics, medical history, and preferences. This personalized approach can enhance user engagement and satisfaction while promoting proactive healthcare management.

- **Enhanced Security Measures:** The system implements robust security measures, including SSL encryption, secure password policies, and CSRF protection, to ensure the confidentiality and integrity of user data. This commitment to security safeguards user privacy and builds trust in the system's reliability and credibility.

- **Scalability and Flexibility:** Designed with scalability in mind, the system can accommodate future growth and expansion, adapting to evolving user needs and technological advancements. Its flexible architecture allows for seamless integration with external systems and services, enabling interoperability and interoperability with other healthcare platforms and solutions.

References

- Published in: 2020 International Conference on Power Electronics & IoT Applications in Renewable Energy and its Control (PARC), Author: Prakhar Srivastava; Nishant Singh, Automatized Medical Chatbot (Medibot): https://ieeexplore.ieee.org/document/9087096

- Published in: 2020 8th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO), Author: Lekha Athota; Vinod Kumar Shukla; Nitin Pandey; Ajay Rana, Chatbot for Healthcare System Using Artificial Intelligence: https://ieeexplore.ieee.org/document/9197833

- Published in: 2019 IEEE 21st Conference on Business Informatics (CBI), Author: Mubashra Akhtar; Julia Neidhardt; Hannes Werthner, The Potential of Chatbots: Analysis of Chatbot Conversations: https://ieeexplore.ieee.org/document/8808076

Marwadi University Marwadi Chandarana Group	Marwadi University Faculty of Engineering and Technology Department of Information and Communication Technology
Semester: 8th	Subject: Project (01CT1801)
A.Y. 2023-24	Date: 10/2/2024

Project review 1

Project title: OnLine Medical Diagnosis system

Name of guide: Prof. HariKesh Chauhan

Name and address of company (if it is industrial project):

Email ID of industrial guide: ~~prathambuddhadev111596@marwadiuniversity.ac.in~~

Enrolment no.	Name of the student	Signature
92110133003	Pratham Buddhadev	Pratham

Members of examination	Param Teraiya (Senior SE, middleware.io)	ONLINE.
	Prof. Nishith Kotak	Kotak

Remarks from examination panel:

- Communication skills need to improve.
- Idea of model used, is not clear.
- Need to check & compare multiple algos. & models
- How features are related / corelated needs to be determined.
- Dynamics of Patients needs to be considered.
- Only current Symptoms are considered. Need to focus on Severity of disease / symptoms, history of Patients, etc.
- Consult doctor & understand the severity of Symptoms.

Marwadi University Marwadi Chandarana Group	Marwadi University Faculty of Engineering and Technology Department of Information and Communication Technology
Semester: 8th	**Subject: Project (01CT1801)**
A.Y. 2023-24	**Date: 09/03/2024**

Project review 2

Project title: OMDS: Online medical Diagnosis System

Name of guide: Prof. Harikesh Chauhan

Name and address of company (if it is industrial project):

—

Email ID of industrial guide: —

Enrolment no.	Name of the student	Signature
92110133003	Pratham Buddhadev	ONLINE.

Members of examination		
	Prof Nishith Kotak	Nkotak

Remarks from examination panel:

→ Implemented on some basic testcases.

→ Not included any synonymous structures like fever & high temp.

→ Doesn't seem to have any severe role of ML, seems to be static only.

→ Refer contextual emb. approaches like SBERT, USE, etc.